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Artificial Intelligence for Interpreting Occupational Health Examination Tests: A Systematized Review of Evidence and Gaps

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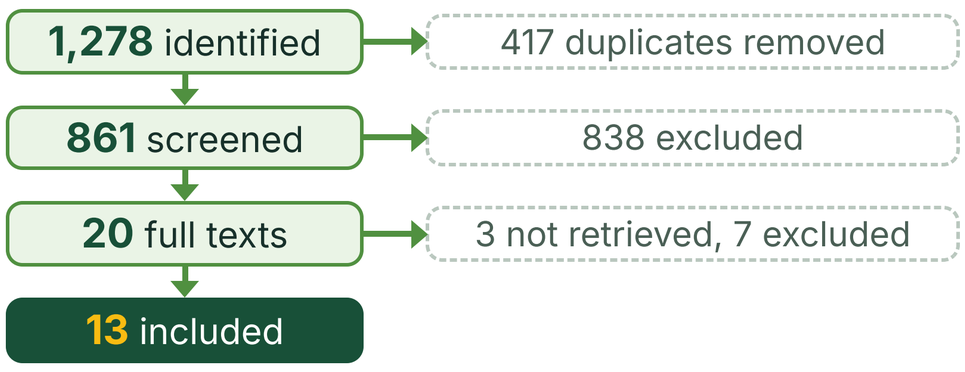
01 Background

- Periodic occupational examinations generate **large volumes of test results**.
- Whether **artificial intelligence (AI)** validly interprets them remains unclear.

Aim: to review the evidence on AI interpreting workers' examination tests against a clinical reference standard.

02 Methods

Design	Systematized review, reported with PRISMA 2020
Databases	PubMed, Scopus, ProQuest, IEEE Xplore
Search	Studies from 2006 onwards
Inclusion	AI interpreting workers' examination tests against a clinical reference standard
Exclusion	Prediction models
Appraisal	JBI diagnostic test accuracy checklist; no pooling



03 Results — Admired behind glass: 0 of 13 evaluated prospectively.

Chest radiography — 9
Reference: expert reading or clinical diagnosis
AUC 0.84–1.00 · Se 62–100% · Sp 46–98%
External validation 3 · Prospective 0

Audiometry — 1
Reference: claimants vs unexposed controls
Se 98.6%, Sp 90.2%
External validation 0 · Prospective 0

Electrocardiography — 1
Reference: examination ECG diagnosis
AUC 0.92–0.98
External validation 0 · Prospective 0

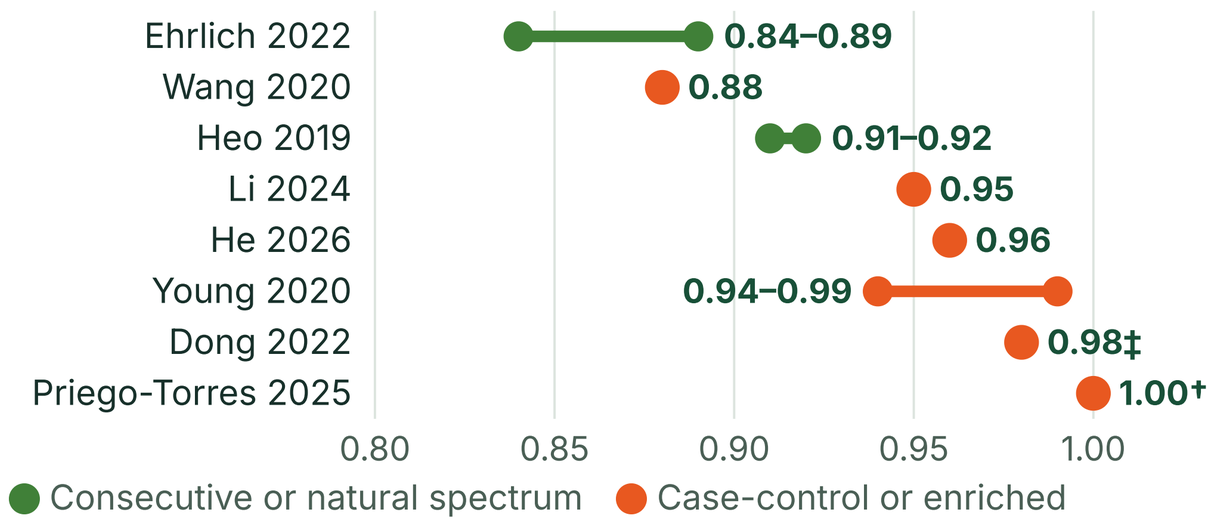


Fitness to work — 0
No study provided AI with the job-demand information required for fitness-to-work decisions

Multimodal records — 2
Reference: physicians' fit/unfit status
AUC 0.60; kappa 1.00 vs one physician rating documents
agreement ≠ accuracy

Spirometry, laboratory — 0
No eligible study: two empty pedestals

Imaging AUC by study design (not pooled)



High AUC ≠ high sensitivity: AUC 0.88 with sensitivity 62%[1]

Case-control and enriched datasets likely overstate screening performance

† Cross-validation, public controls. ‡ Region-level features.

Quality and validation

JBI DIAGNOSTIC TEST ACCURACY CHECKLIST	YES
Q1 Consecutive/random	2
Q2 Case-control avoided	5
Q3 Exclusions appropriate	1
Q4 Index test blinded	11
Q5 Threshold pre-specified	1
Q6 Appropriate reference	6
Q7 Reference blinded	11
Q8 Appropriate interval	12
Q9 Same reference for all	10
Q10 All analysed	7

Yes · Unclear · No · Not applicable

DATA EXTRACTION	YES
External validation	3
NPV reported	2
Prospective evaluation	0

⚠ Limitations of the current evidence

- Evidence centres on chest radiography (9 of 13 studies).
- Clinical-setting spirometry and ECG AI performs well but is unvalidated in workers.[2,3]
- Reference standards are imperfect: human readers agreed only moderately (kappa 0.42–0.66).[1,4]
- One system was deployed in more than 10 hospitals without reported implementation outcomes.[5]

04 Conclusion

Evidence is insufficient to support AI interpretation in periodic examinations. AI is best positioned as decision support under physician responsibility.

Adoption requires:

External validation · **Prevalence-specific NPV** · **Comparison with occupational physicians** · **Prospective evaluation**

Scan: PRISMA flow, 13 included studies, JBI appraisal

References

1. Wang X, Yu J, Zhu Q, Li S, Zhao Z, Yang B, et al. Potential of deep learning in assessing pneumoconiosis depicted on digital chest radiography. Occup Environ Med. 2020;77(9):597-602.
2. Topalovic M, Das N, Burgel PR, Daenen M, Derom E, Haenebalcke C, et al. Artificial intelligence outperforms pulmonologists in the interpretation of pulmonary function tests. Eur Respir J. 2019;53(4):1801660.
3. Attia ZI, Kapa S, Lopez-Jimenez F, McKie PM, Ladewig DJ, Satam G, et al. Screening for cardiac contractile dysfunction using an artificial intelligence-enabled electrocardiogram. Nat Med. 2019;25(1):70-4.
4. Ehrlich R, Barker S, Te Water Naude J, Rees D, Kistnasamy B, Naidoo J, et al. Accuracy of computer-aided detection of occupational lung disease: silicosis and pulmonary tuberculosis in ex-miners from the South African gold mines. Int J Environ Res Public Health. 2022;19(19):12402.
5. Li X, Xu M, Yan Z, Xia F, Li S, Zhang Y, et al. Deep convolutional network-based chest radiographs screening model for pneumoconiosis. Front Med (Lausanne). 2024;11:1290729.